



PREDICTING LOAN DEFAULT USING RANDOM FOREST PROPOSAL

PROBLEM STATEMENT

Financial institutions today face challenges in borrower defaults. Companies such as the Lending Club attempt to bridge the gap for willing investors to invest in borrowers seeking alternatives to traditional banking. These peer lending companies still face significant risks due to defaults. This project aims to accurately predict whether a borrower will default.

DATASET

The dataset contains loan data from the Lending Club, a peer-to-peer lending company that matches investors to borrowers. This dataset was obtained from Kaggle and covers loans issued from 2007-2015.

MODEL/METHOD

A Random Forest classifier will be utilized to predict whether a borrower will default on a loan. Random Forest combines multiple decision trees to improve predictive accuracy. A baseline model will be used for comparison.

SUMMARY

The goal is to develop a predictive model that estimates the likelihood of borrower default. Accurately assessing credit risk helps institutions make better lending decisions. This model will use characteristics such as income, employment, and credit to predict default.

PREPARED BY:

La Mia Pruitt | Senior

PREPARED FOR:

Nhu Nguyen | Professor